Mol Biol Rep
DOI 10.1007/s11033-013-2627-y

Differences of DNA methylation profiles between monozygotic twins' blood samples

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Received: 17 July 2012 / Accepted: 30 April 2013
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Abstract Monozygotic twins (MZs) share an identical genomic sequence, which makes it impossible to discriminate one another with conventional genetic markers like STRs. On the other hand, phenotypic discordance between MZs implies the existence of different epigenetic characteristics. DNA methylation, an essential epigenetic modification, however, might be a potential biomarker to solve the forensic puzzle. In this study, we examined 22 pairs of MZs with a methylation BeadChip including 27,578 CpG sites. The results suggested that MZs exhibited remarkable differences of genome-wide 5-methylcytosine. According to a set of criteria of selection, 92 CpG sites with significant differences of methylation status within MZs were identified from the global epigenome. In conclusion, this pilot study suggested that CpG methylation profile could be a useful biomarker in individual identification of MZs.

Keywords Forensic genetics · Individual identification · Monozygotic twins · DNA methylation

Electronic supplementary material The online version of this article (doi:10.1007/s11033-013-2627-y) contains supplementary material, which is available to authorized users.

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Introduction

Theoretically, monozygotic twins (MZs) share completely identical genomic DNA sequence, which makes it impossible to distinguish MZs from one another with conventional DNA genetic markers [1, 2], like short tandem repeats (STRs), single nucleotide polymorphisms (SNPs) or Insertion/Deletion polymorphisms (InDels). However, MZs almost always show various degrees of phenotypic discordance [3–5]. In the last decade, evidences have been accumulating that epigenetic modifications of DNA and histones can play a primary role in phenotypic outcomes, including human diseases [6–10]. Such results provided an unprecedented idea about how to discriminate MZs for forensic purposes. As for the application of forensic genetics, DNA methylation might be one of the most promising epigenetic markers in the discrimination of MZs. Previous studies have showed that there are DNA methylation differences within MZs [6, 7]. Besides, high-performance methods of site-specific or epigenome-wide DNA methylation detection, such as high resolution melting curve (HRM), MethyLight and bead arrays [11–13] had been established. In this study, an epigenome-wide scan was employed to identify those potential CpG sites for discrimination of MZs.

Materials and methods

Sample collection

Blood samples were taken from 13 pairs of female and 9 pairs of male MZs respectively. Their age ranges from 17 to 74 years old. All participants signed the informed consent forms, and the study was approved by the local ethics

Published online: 07 May 2013

committee. Genomic DNA was extracted using a QIAamp DNA blood mini kit based on the manufacturer's instructions (Qiagen, Hilden, Germany).

Determination of monozygosity

All the genomic DNA samples of 22 pairs of MZs were genotyped with AmpFlSTR Identifier kit (Applied Biosystems, Foster City, CA, USA) and an in-house Insertion/Deletion polymorphisms (InDel) genotyping system including 30 InDel loci (InDel_typer30 system) [14, 15] to determine their monozygosities. PCR amplification reactions were applied using a GeneAmp PCR system 9700 (Applied biosystems, Foster city, CA, USA). The amplified products were separated by capillary electrophoresis on ABI PRISM 3130 Genetic Analyzers (Applied Biosystems, Foster City, CA, USA) according to the manufacturer's recommended protocols. The data were analyzed with an allelic ladder, a positive control and a negative control using GeneMapper ID software version 3.2 (Applied biosystems, Foster city, CA, USA).

Bisulfite treatment

Genomic DNA was bisulfite modified by EZ DNA methylation-Glod kit (Zymo Research, Orange, CA, USA) according to the manufacturer's suggested protocols for the Infinium Methylation Assay. Bisulfite sodium converts unmethylated cytosine to uracil. The bisulfite-modified DNA was stored at −20 °C and used within a week for analysis.

Illumina Infinium human Methylation27 Beadchip assays

Bisulfite-converted genomic DNA was analyzed using Illumina's Infinium human Methylation27 Beadchip kit (performed at Shanghai Biostar Genechip, Inc., Shanghai, Peoples Republic of China). The Beadchip contains 27,578 CpG dinucleotides sites, encompassing 14,473 well-annotated and unique human RefSeq genes. BeadChip process was performed with reagents provided in the kit. The data was analyzed following the manufacturer's manual. In short, 4 μl of bisulphite-converted genomic DNA was denatured in 0.014 N sodium hydroxide, neutralized and amplified with kit-provided reagents. Then it was buffered for 20–24 h at 37 °C. After being fragmented, 12 μl of each sample was loaded onto a 12-sample chip. Next, the chips were assembled into a hybridization chamber as instructed in the manual. After incubation at 48 °C for 16–20 h, the chips were washed with washing buffers. Then they were assembled and placed into a fluid flow-through station for primer extension reaction and stained with reagents and buffers provided in the kit. The images of polymer-coated chips were processed in Illumina's iScan scanner. Data were put into GenomeStudio software for further analysis.

Data analysis

Background normalization was conducted using the negative control signals of each sample. Average normalization was done to minimize chip-to-chip variation. The average intensity values of the first color channel for all the wells in each chip were used to calculate the mean value, which were scaled to 1. This procedure ensures that each chip has the same average intensity in each color channel. Methylation status for each CpG site then can be showed as a beta (β)-value, ranging from 0 (completely unmethylated) to 1 (completely methylated). The beta (β)-value was defined as a ratio of methylated-probe signal intensity to total signal intensity of methylated and unmethylated probes.

The quality of the BeadChip data was assessed through two different ways. One was the linear regression of the correlation of two pairs of technical replicates, and the other was the evaluation of DNA methylation stability of house-keeping genes in different samples.

In order to evaluate the difference of methylation level on a specified CpG site within each pair of MZs, a scoring system, "Diffscore", was applied according to the manufacturer's recommendation. DiffScore is an important parameter in the

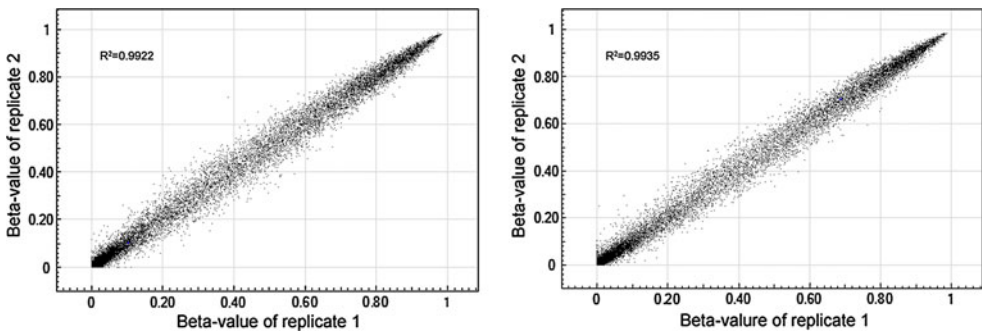


Fig. 1 Evaluation of reproducibility of the Beadchip assay data. The panels demonstrated the reproducibility of replicated assays on bisulfite genomic DNA derived from one female (*left*) and one male (*right*) sample, respectively

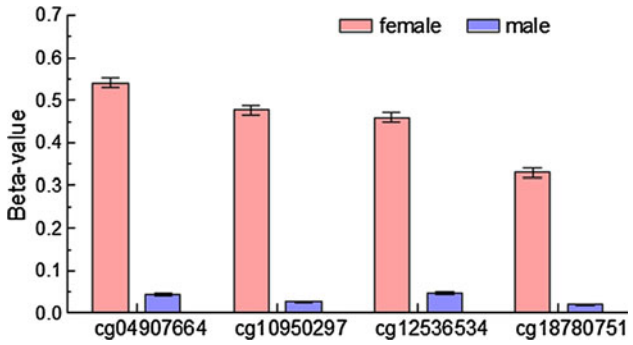


Fig. 2 Evaluation of stability of the Beadchip assay data. The average β-values of the four CpG sites on the two X-linked house-keeping genes, *EFNB1* (cg04907664 and cg10950297) and *G6PD* (cg12536534 and cg18780751), showed their approximate hemi-methylation in female samples and hypomethylation in male samples

analysis of the Infinium human Methylation27 Beadchip data with Genomestudio software, a methylation analysis tool provided by Illumina Inc. The DiffScore value on a specified CpG site between a pair of sample (A and B) can be calculated with a formula (DiffScore $= 10 \times [\mathrm{sgn}(\beta_{A-}\beta_B)] \times \log_{10}P$) according to the Illumina Custom Algorithm. For a P value of 0.05, DiffScore equals ± 13. The Illumina methylation assay is reported to be accurate for β value differences larger or equals to 0.17 [13], which can thus be chosen as the threshold for methylation differences within MZs on a given CpG site. In order to determine the list of significantly different methylation CpG sites, two criteria were applied in our study: (i) absolute value of DiffScore (ABDs) ≥ 13, as determined by comparative marker-selection analysis (with use of a two-sided t test), and (ii) mean of the absolute value of delta beta ($\Delta\beta = \beta_{A-}\beta_B$) within MZs (ABDβ) ≥ 0.17. Volcano plots of methylation difference within MZs were performed with *R* software (www.r-project.org).

Results and discussion

Determination of monozygosity

STR and InDel profiles of each sample were determined and compared to the corresponding siblings. When no differences could be detected in the 15 STR loci provided by the Identifiler Kit and in the 30 InDel markers provided by InDel_typer30 system, the twins were considered as being MZs. The total discrimination power (TDP) of the 15 STR loci was 0.999999999999999998 and that of the 30 InDel loci was 0.99999999999841 according to the allele frequency database published by Li et al. [15, 16]. The combined discrimination power of the two different multiplex genotyping systems is high enough in Chinese Han population that it could be assumed that the results are true and reliable. In all cases, the true MZ twin status was confirmed.

Reproducibility and stability of the high-throughput Illumina Beadchip data

Reliability of the Beadchip data is the prerequisite to find out differential methylation CpG sites within MZs, which had been well evaluated in reported study [13]. The same strategies were applied in our study. For the reproducibility testing, the bisulfite conversion genomic DNA of one female and one male sample were tested twice, respectively. Highly reproducible DNA methylation profiles between these technical replicates were obtained (Fig. 1) with a R^2 of 0.9922 in female and 0.9935 in male samples, when β-values were compared with each other.

In order to evaluate the stability of the Beadchip data, β-values of two commonly used house-keeping genes, *ACTB* and *GAPDH*, and two X-linked house-keeping genes,

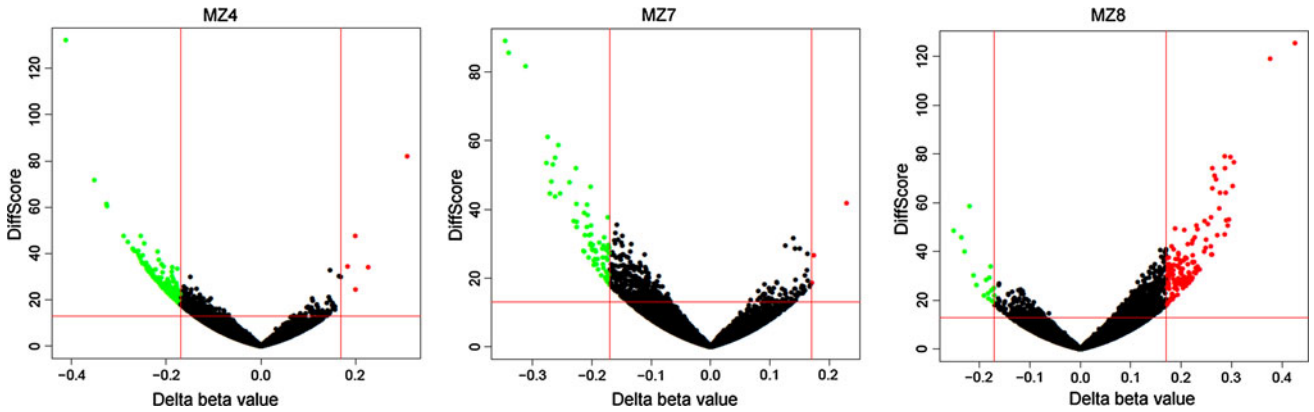


Fig. 3 Volcano plots of three pairs of MZs at different age (years). The X axis represents the $\Delta\beta$ value which indicates the relative methylation change within MZs. The Y axis represents the ABDs value from paired t test. Red dot represents the $\Delta\beta$ value larger than 0.17 while ABDs larger than 13. Green dot represents the $\Delta\beta$ value smaller than -0.17 while ABDs larger than 13. The *left*, *center* and *right* panel indicate the volcano plots of MZ4 (age = 23), MZ7 (age = 50) and MZ8 (age = 74), respectively

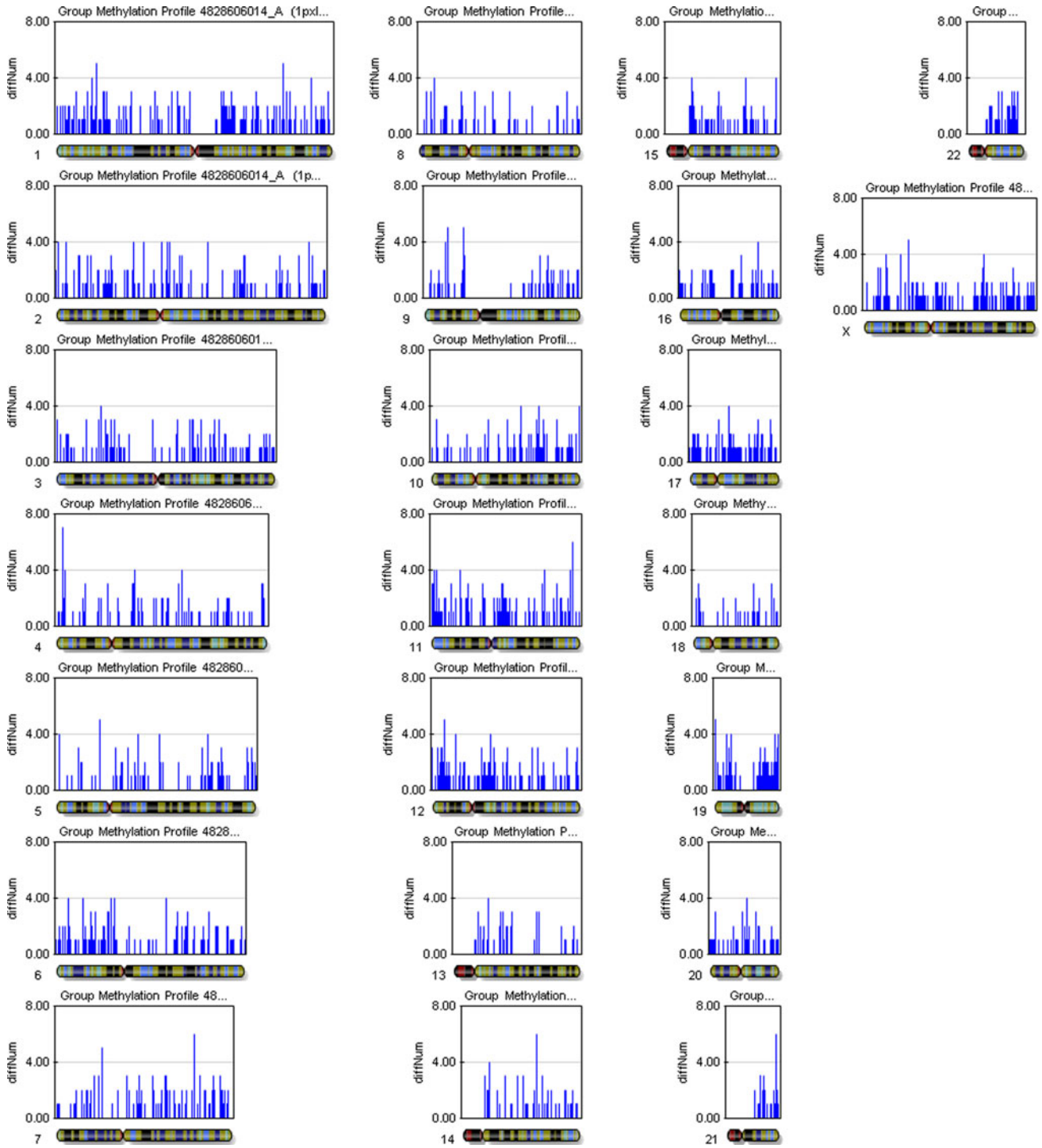


Fig. 4 A chromosomal karyogram depicting the frequency of differentially methylated CpG site among all 22 pairs of MZs. Blue bars represent those CpG sites which meet the criteria of ABDs $\geq$ 13 and ABDβ $\geq$ 0.17. Height of blue bar represent times of the CpG site observed among all the 22 pairs of MZs (diffNum). When diffNum $\geq$ 4, $f_{\mathrm{dm}} \geq 0.1818$. Human Mitochondria genome was indicated as "M" in the Figure

EFNB1 and *G6PD,* were analyzed for all the 44 samples. In normal cells, *ACTB* and *GAPDH* were usually hypomethylated, and were usually used as "negative" reference in methylation detection of a specific target gene [17, 18]. Our results showed that the average β-values of the two CpG sites from *ACTB* (TargetID is cg23175281 and cg23261233 in the BeadChip) were 0.01 and 0.02, respectively. The average β-values of the two CpG sites from *GAPDH*

(cg00241355 and cg20917484) were 0.02 and 0.03, respectively. In general, methylation levels of X-linked house-keeping genes correlated well with the gender of the sample source, i.e., no or very low methylation was detected in male somatic cells while approximate hemi-methylation was detected in female somatic cells. This is because DNA methylation is the main mechanism of compensation for the dosage of functional X-linked genes between male and female people [19]. Methylation status of the 4 CpG sites from *EFNB1* and *G6PD* in female and male samples was presented in Fig. 2. Clearly, our results were concordant with the theory mentioned above. In short, all of the results indicated the stability of the Beadchip used in our study. Combined with its high reproducibility, it could be ensured that the Beadchip technology is powerful in identifying actual DNA methylation differences between MZs, rather than mere artificial differences due to technical variation.

DNA methylation differences between MZs

Several studies have been conducted to identify epigenetic differences between MZs, focusing on either the selected genes or in the overall epigenome. Most of these studies were interested in the molecular mechanisms of phenotypic discordance within MZ pairs, especially human diseases related [6–10]. These studies intensively hinted the potential utilization of epigenetic markers, such as CpG sites in forensic genetics. In our study, however, differences of DNA methylation profiles of each pair of the 22 MZs were illustrated with volcano plots according to the criteria of ABDs $\geq$ 13 and ABD$\beta \geq 0.17$. Volcano plots of three pairs of MZs at different age were presented at Fig. 3. These volcano plots revealed the CpG sites with differential methylation status within each pair of the 22 MZs. Consequently, 3 616 candidate CpG sites with differential methylation status were determined. Each candidate CpG site meets the two criteria in at least one pair of MZs. However, the most important thing was to pick out those CpG sites which were significantly different among most of the 22 MZs. Frequency of differential methylation status (f_{dm}) of a given CpG site, a new parameter, was defined in our study. It can be used to interpret the times that were observed among all the 22 MZs of the CpG site with differential methylation status within one pair of MZs. The resulting number was called as diffNum. For example, if a CpG site with differential methylation status could be observed in 4 (diffNum = 4) out of the 22 pairs of MZs, the f_{dm} of the CpG site will be 0.1818 (4/22). The larger f_{dm}, the better the CpG site can be used as a marker to discriminate MZs. In our study, the threshold of f_{dm} was artificially specified as $\geq$0.1818 (Table S1).

According to the criterion of f_{dm}, a candidate pool including 92 "best" CpG sites were picked out from the 3 616 candidate CpG sites, which were scattered on 20 autosomes (except chromosome 18 and 22) and X-chromosome (Fig 4). each MZs have been distinguished according to the 92 candidate CpG sites.

Results of previous studies had indicated that differences in the degree of methylation in monozygotic twins may be related to the environment, disease, the location of CpG in the genome and other factors [10, 20]. In this study, the sample type used is peripheral blood. The functions of peripheral blood leukocytes have a great relationship with the immune. Therefore, the authors speculate that the majority of 92 candidate CpG sites may be associated with immune-related.

The initial work presented in this article indicates the potential value of CpG methylation in distinguishing MZs from one another. Furthermore, a definite candidate pool of CpG sites was also determined in our study. However, we've realized that there is a long way to go before we can utilize the primary results in forensic caseworks. This is because major results of this pilot study should be replicated in a larger cohorts study and the following key issues must be scientifically clarified. The first one is the stability of DNA methylation over time as DNA methylation was merely a kind of metastable marker [10, 20–22]. The second one is the specificity of DNA methylation across tissues. It must be emphasized that the candidate CpG sites for distinguishing MZs from one another in this study were determined based on blood samples. Therefore, a reasonable hypothesis might be that, these candidate CpG sites would not be suitable to distinguish MZs from one another using other kinds of forensic samples like buccal swab and semen stain. Finally, a reliable and feasible DNA methylation detection system for those candidate CpG sites must be developed for the convenience of utilization in future forensic casework.

Acknowledgments We are grateful to all volunteers who provided samples for this study. This study was supported by grants from National Key Technology R&D Program of Ministry of Science and Technology of the People's Republic of China (2012BAK16B01) and the National Nature Science Foundation, People's Republic of China (nos. 81222041 and 81172908).

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